Adaptive Spectral Thresholding via Kurtosis-Guided Squared Envelope Analysis for Load-Invariant Bearing Fault Detection

Member 1, Member 2, Member 3, and Member 4

***Abstract*—Rolling element bearing diagnostics rely on envelope spectrum analysis to extract the periodic impulse rates produced by localized surface defects. In industrial operation, however, variable mechanical load alters the shaft slip, the structural damping and the broadband noise floor, so that fixed bandpass filters and static spectral thresholds exhibit high false-alarm rates under heavy load and missed detections under light load. This paper presents a training-free pure Digital Signal Processing (DSP) framework that combines Kurtosis-guided resonance demodulation with local noise-floor adaptive thresholding in the Squared Envelope Spectrum (SES). Cascade second-order-section (SOS) zero-phase IIR filtering, discrete Hilbert transform demodulation, Hann-windowed Fast Fourier Transform (FFT) and order-statistic Median–Median Absolute Deviation (Median–MAD) Constant False Alarm Rate (CFAR) thresholding allow the detector to track load-induced noise-floor shifts and shaft slip automatically. The processing chain is mathematically interpretable, requires no training data, and is sized for fixed-point embedded implementation; it is applied here to the multi-load CWRU and MFPT bearing benchmarks.**

***Index Terms*—Bearing fault diagnosis, Squared Envelope Spectrum (SES), Hilbert transform, spectral kurtosis, IIR/FIR filter design, CFAR, Median–MAD adaptive thresholding.**

I. Introduction

ROLLING element bearings support rotating shafts in motors, pumps, turbines, gearboxes and many other machines. Their condition matters for machine availability, because localized surface damage can develop into severe mechanical failure when it is not found early. A defect on an inner race, outer race or rolling element produces a short-duration contact disturbance each time the rolling contact passes the damaged region. These impacts excite resonances in the bearing and in the supporting structure, and an accelerometer mounted on or near the housing observes them as vibration. The diagnostic information sits in the *repetition rate* of the impacts, and it is recovered by demodulating the resonance that carries them, which is the principle behind envelope analysis [1], [2].

The measured vibration is not stationary across operating conditions. Load changes the contact force and the strength of the defect-induced impacts, and it also changes the overall vibration and noise floor. Effective shaft speed differs between load conditions, and local slip makes the actual impact repetition rate depart slightly from the ideal value predicted by bearing geometry. A fixed threshold on raw amplitude or on the spectrum is therefore not expected to behave consistently at all loads.

The approach taken here is a pure DSP framework, not a model trained on labelled examples. Every stage can be inspected on its own: the characteristic frequencies, the filters, the transforms and the detection thresholds are all explicit. Interpretability of this kind does not guarantee generalization. Bearing geometry, shaft speed, sensor placement and validation data are still required.

This paper applies two classical operations where they matter most. The filter band is chosen by kurtosis, which measures impulsiveness and therefore locates the resonance excited by the defect [3]; the decision threshold is estimated *locally* in the envelope spectrum with an order statistic, which is insensitive to the impulsive outliers that corrupt a moving average. The resulting detector contains no learned parameters and can be implemented on a fixed-point microcontroller.

The contributions of this paper are:

- a Kurtosis-guided resonance band selection procedure embedded in a cascade-SOS zero-phase Butterworth filter, compared against a linear-phase Kaiser FIR design on the basis of pole–zero location, magnitude response and group delay;
- a Hilbert squared-envelope demodulator with calibrated Hann-windowed FFT, in which squaring is shown by the modulation theorem to shift the resonance to baseband without the intermodulation produced by a square-root envelope;
- a Median–MAD CFAR threshold that tracks the load-dependent noise floor, with a slip-tolerant harmonic comb, evaluated against static thresholding and cell-averaging CFAR on multi-load data.

II. Signal Model and Theoretical DSP Framework

A. Continuous-to-Discrete Fault Signal Model

A localized bearing defect can be modelled as a sequence of short impacts. Let the impact times be $t_m = m/f_{\text{fault}} + \varepsilon_m$, where f_{fault} is the nominal repetition rate and ε_m is a timing perturbation caused, for example, by local slip or speed fluctuation. Each impact has amplitude A_m and excites the mechanical structure with impulse response $h(t)$; background machine vibration and sensor noise enter as $w(t)$:

$$x(t) = \sum_m A_m \, h(t - t_m) + w(t). \tag{1}$$

Manuscript submitted for the ED5017 Digital Signal Processing Mini-Project. The authors are with the Department of Engineering.

For an underdamped structural mode the causal impulse response is a decaying sinusoid,

$$h(t) = e^{-\sigma t}\cos(2\pi f_r t)\,u(t), \qquad t \geq 0, \tag{2}$$

where σ is the exponential decay rate, f_r the structural resonance frequency and $u(t)$ the unit step function. The response is causal because it begins at the impact time and decays afterwards. This repeated-impact, exponentially decaying representation is the classical localized-defect model [4].

Sampling with period $T_s = 1/f_s$ gives

$$x[n] = \sum_m A_m\, h[n - n_m] + w[n], \qquad n_m \approx \text{round}(f_s t_m). \tag{3}$$

Equivalently, the impact sequence is $p[n] = \sum_m A_m \delta[n - n_m]$, and the measured vibration is $x[n] = p[n] * h[n] + w[n]$, with $*$ the linear convolution. The nominal interval between impacts, in samples, is $N_p \approx f_s/f_{\text{fault}}$. In practice N_p need not be an integer, and small timing variations shift the observed spectral peaks away from the ideal kinematic frequencies.

The sampling frequency must exceed twice the highest vibration frequency to be represented, $f_s > 2f_{\max}$, and the acquisition chain must suppress out-of-band components before sampling to prevent aliasing. The records used here cover 12 and 48 kHz (CWRU) and 48.828 and 97.656 kHz (MFPT), with Nyquist frequencies of 6, 24, 24.414 and 48.828 kHz, respectively.

In the z-plane, $h[n]$ corresponds to a complex-conjugate pole pair

$$z_{1,2} = re^{\pm j\omega_r}, \qquad r = e^{-\sigma T_s} < 1, \tag{4}$$

so stability follows from $|z_{1,2}| < 1$. The model makes the central point of the paper explicit: the fault signature A_m is a *low-frequency modulation* of a *high-frequency carrier* ω_r, and the two are separated by demodulation rather than by direct spectral inspection.

B. Pre-Processing and Kurtosis-Guided Bandpass Filter Design

1) DC Offset Removal and Detrending: Accelerometers come with a DC offset. The sensor electronics and the signal conditioning hardware add a constant bias, and it appears as a large peak at 0 Hz. Its sidelobes then leak into the low-frequency bins that matter. Before any filtering, the signal $x[n]$ of length N is centered by subtracting its sample mean:

$$x_0[n] = x[n] - \mu_x, \quad \text{where} \quad \mu_x = \frac{1}{N}\sum_{m=0}^{N-1} x[m] \tag{5}$$

In the frequency domain this removes the $\omega = 0$ term of the DTFT:

$$X_0(e^{j0}) = \sum_{n=0}^{N-1} x_0[n] = 0 \tag{6}$$

After this step the filter bank sees only the vibration riding on a zero baseline. The subtraction is exact in the DFT sense: it changes the bin at $\omega = 0$ and leaves every other coefficient untouched.

2) Sub-Band Kurtosis Resonance Selection: Below about 1000 Hz the spectrum is crowded. Shaft harmonics, structural vibration and load-dependent friction noise all sit there, and they hide the bearing impulses. The impulses themselves are short impacts. Each time a rolling element hits a defect on the raceway, it excites a high-frequency resonance of the housing. That band carries sharp bursts. The low-frequency interference is stationary and does not. A resonance belongs to the structure, not to the fault: it rings at about the same frequency after every impact, which is why one band can collect many bursts at once.

A fixed bandpass filter cannot find the resonance reliably, because the frequency depends on the machine and on how it is mounted. The band has to be picked per record. A filter bank built on the Fast Kurtogram [3] does this without training data. The Nyquist interval $[0, f_s/2]$ is split at several dyadic and 1/3-binary levels. On a 12 kHz record the limit is 6000 Hz, so the dyadic levels give bands of 3000, 1500, 750 Hz and so on, while the 1/3-binary levels add widths of 2000, 1000, 500 Hz. The 2000 Hz band used in Section IV comes from that second set. At level k there are $M_k = \text{round}(2^k)$ bands, and band i has bandwidth and center frequency

$$\Delta f_k = \frac{f_s}{2M_k}, \quad f_{c,i} = \left(i + \frac{1}{2}\right)\Delta f_k \tag{7}$$

For each band output $y_{k,i}[n]$, impulsiveness is measured by the fourth standardized central moment, the Kurtosis [5]:

$$K(f_{c,i}, \Delta f_k) = \frac{\frac{1}{N}\sum_{n=0}^{N-1}\left(y_{k,i}[n] - \mu_y\right)^4}{\left(\frac{1}{N}\sum_{n=0}^{N-1}\left(y_{k,i}[n] - \mu_y\right)^2\right)^2} \tag{8}$$

with μ_y the mean of the band signal. The fourth power in the numerator is what makes spikes count: one large transient outweighs many small samples. The denominator, the squared variance σ_y^4, normalizes out the band energy, so a loud band does not win by being loud. Gaussian noise gives $K \approx 3$. A band full of repeated impacts gives K well above 3. The band with the largest K is selected:

$$\{f_c^*, BW^*\} = \arg\max_{f_{c,i},\,\Delta f_k} K(f_{c,i}, \Delta f_k) \tag{9}$$

Its cutoffs are $f_{\text{low}} = f_c^* - BW^*/2$ and $f_{\text{high}} = f_c^* + BW^*/2$, that is $\omega_1 = 2\pi f_{\text{low}}/f_s$ and $\omega_2 = 2\pi f_{\text{high}}/f_s$.

3) Bandpass Filter Design: Kaiser FIR versus Butterworth IIR: Two filters are designed for that passband, a linear-phase FIR and a Butterworth IIR. Both have to reject out-of-band motor interference and broadband noise.

The FIR is Type-I, with even order M and $L = M + 1$ taps. The ideal bandpass impulse response is truncated around its symmetry center $\alpha = M/2$ and multiplied by a Kaiser window [6]:

$$h_{\text{FIR}}[n] = \left(\frac{\sin(\omega_2(n-\alpha)) - \sin(\omega_1(n-\alpha))}{\pi(n-\alpha)}\right) \times \frac{I_0\left(\beta\sqrt{1 - \left(\frac{n-\alpha}{\alpha}\right)^2}\right)}{I_0(\beta)} \tag{10}$$

for $0 \le n \le M$. $I_0(\cdot)$ is the zeroth-order modified Bessel function of the first kind, and β sets the trade between transition width and sidelobe attenuation. The taps are symmetric, $h_{\mathrm{FIR}}[n] = h_{\mathrm{FIR}}[M-n]$, so the phase is exactly linear and the delay is flat:

$$\tau_{g,\mathrm{FIR}}(\omega) = -\frac{d\angle H_{\mathrm{FIR}}(e^{j\omega})}{d\omega} = \frac{M}{2} \text{ samples} \tag{11}$$

An FIR of this form has all its poles at $z = 0$ and only zeros elsewhere, so it is stable whatever the coefficients do. The price is arithmetic: a longer window buys a steeper transition, and every extra tap costs one more operation per sample. The design in Section IV uses $\beta = 8$, which targets roughly -80 dB of stopband attenuation.

The second design is a 4th-order Butterworth bandpass, obtained with the bilinear transform [7]. That transform maps the left half of the s-plane into the unit circle:

$$s = \frac{2}{T_s}\left(\frac{1-z^{-1}}{1+z^{-1}}\right) \tag{12}$$

with $T_s = 1/f_s$. The mapping compresses frequency nonlinearly, so the digital cutoffs are pre-warped before the prototype is transformed:

$$\Omega_k = \frac{2}{T_s}\tan\!\left(\frac{\omega_k}{2}\right), \quad k \in \{1,2\} \tag{13}$$

After pre-warping, the -3 dB points land exactly on f_{low} and f_{high}, and the passband stays maximally flat.

4) Cascade SOS Realization and Zero-Phase Filtering: A 4th-order bandpass has eight poles. Written in direct form, that is a degree-eight denominator, and in fixed point a small error in one coefficient moves the poles a long way. They can cross the unit circle. The fix is to factor the transfer function into $K = 4$ second-order sections:

$$H_{\mathrm{IIR}}(z) = G\prod_{k=1}^{K} H_k(z) = G\prod_{k=1}^{K}\frac{b_{0k}+b_{1k}z^{-1}+b_{2k}z^{-2}}{1+a_{1k}z^{-1}+a_{2k}z^{-2}} \tag{14}$$

where G is the overall gain. Each section holds one conjugate pair $p_k, p_k^* = r_k e^{\pm j\theta_k}$. Stability needs the ROC $|z| > r_{\max}$ to contain the unit circle, so every pole radius must satisfy

$$r_k = \sqrt{a_{2k}} < 1, \quad \forall k \in \{1,2,\ldots,K\} \tag{15}$$

The cascade is cheap to run, but its phase is not linear near the band edges. Group delay then varies with frequency, which smears a short impact in time and lowers its peak. Forward-backward filtering removes the problem. The record is filtered, reversed, filtered again, and reversed back:

$$Y_{\mathrm{BP}}(z) = H(z)H(z^{-1})X_0(z) \tag{16}$$

On the unit circle, $H(e^{-j\omega}) = H^*(e^{j\omega})$ for real coefficients, so

$$H_{\text{zero-phase}}(e^{j\omega}) = H(e^{j\omega})H^*(e^{j\omega}) = |H(e^{j\omega})|^2 \tag{17}$$

That response is real and non-negative, so the phase is zero and the group delay is zero samples. The impacts keep their time positions and their shape, and the stopband attenuation doubles in decibels. The passband edge shifts a little inward, because the effective filter is the square of the one that was designed.

C. Hilbert Demodulation and Windowed Squared Envelope Spectrum

The bandpass stage leaves $y_{\mathrm{BP}}[n]$ as a series of short bursts of structural ringing. The bursts repeat at the fault rate f_{fault}. The FFT of $y_{\mathrm{BP}}[n]$ puts the energy around the resonance ω_r, so the fault rate is almost invisible in the spectrum. It has to be moved down to a low-frequency region, where the harmonics are far apart and easy to separate. That operation is demodulation.

The first step is the analytic signal, built with the Discrete Hilbert Transform [6]:

$$z[n] = y_{\mathrm{BP}}[n] + j\hat{y}_{\mathrm{BP}}[n], \tag{18}$$

where $\hat{y}_{\mathrm{BP}}[n] = \mathcal{H}\{y_{\mathrm{BP}}[n]\}$ is the Hilbert transform of the filtered signal. A real record has a spectrum on both sides of 0 Hz, and the two halves mirror each other. The transform keeps the positive half and drops the rest, which is what makes the envelope well defined. It works as a 90° phase shifter,

$$H_{\mathrm{Hilbert}}(e^{j\omega}) = -j\ \mathrm{sgn}(\omega), \tag{19}$$

so every positive frequency is shifted by $-\pi/2$ and the negative frequencies are removed.

The usual envelope takes a square root of $y_{\mathrm{BP}}^2 + \hat{y}_{\mathrm{BP}}^2$. The squared envelope is used here instead:

$$e[n] = |z[n]|^2 = z[n]z^*[n] = y_{\mathrm{BP}}^2[n] + \hat{y}_{\mathrm{BP}}^2[n]. \tag{20}$$

Squaring does two useful things. A square in time is an auto-convolution in frequency, so the modulation theorem moves the carrier ω_r down to baseband (0 Hz) without distortion. A square root does not behave this way. It is non-linear, and the intermodulation harmonics it adds can hide a weak fault. Squaring also favours large amplitudes: an impact peak is squared into a much larger number, while the background changes little. The fault-to-noise ratio therefore improves before any threshold is applied.

The squared envelope is never negative, so it carries a large DC term. That term is much bigger than the fault harmonics, and its mainlobe leaks into the low-frequency bins where the cage and shaft harmonics sit. The mean is removed first:

$$e_0[n] = e[n] - \mu_e, \qquad \mu_e = \frac{1}{N}\sum_{n=0}^{N-1} e[n]. \tag{21}$$

A fault frequency rarely sits exactly on a DFT bin centre. At 107.36 Hz, with $\Delta f = f_s/N_{\mathrm{FFT}}$, the mismatch is unavoidable. Rectangular truncation then spreads sidelobes into the neighbouring bins. An N-point Hann window controls them:

$$w[n] = \tfrac{1}{2}\left(1 - \cos\frac{2\pi n}{N-1}\right), \qquad n = 0,1,\ldots,N-1. \tag{22}$$

The window brings both ends to zero, which halves the coherent amplitude. It also widens the mainlobe, so a little frequency resolution is lost, and that trade is accepted for the much lower sidelobes. The coherent gain

$$CG = \frac{1}{N}\sum_{n=0}^{N-1} w[n] = \tfrac{1}{2} \tag{23}$$

puts the amplitude back, so the SES levels keep their physical meaning. The one-sided SES is then computed with an N_{FFT}-point radix-2 FFT:

$$\mathrm{SES}[k] = \frac{2}{N \cdot CG} \left| \sum_{n=0}^{N-1} w[n]\, e_0[n]\, e^{-j\frac{2\pi}{N_{\mathrm{FFT}}}kn} \right|, \qquad (24)$$

for $k = 0, 1, \ldots, N_{\mathrm{FFT}}/2$, where bin k corresponds to $f_k = k\, f_s/N_{\mathrm{FFT}}$ Hz.

D. Local Noise-Floor Adaptive Spectral Thresholding

Load changes the SES floor in both directions. At 0 HP the rolling elements press lightly on the raceways. The impacts are weak, and a fixed threshold set for normal running sits above the fault peaks, so they are missed. At 3 HP friction raises the whole floor, and the same threshold is crossed by noise in dozens of bins. No single fixed value satisfies both cases.

A CFAR rule [8] handles both cases. It slides a window along the SES and estimates the local floor from the bins around the test bin, so the comparison is always made against the floor that is actually there. No training data is needed, and the false-alarm rate stays fixed as the load changes.

For a test bin k, the N_G bins on each side are kept out of the estimate. The Hann mainlobe is four bins wide, so $N_G = 4$ is enough to stop a fault peak from entering its own estimate. The $N_R = 25$ bins beyond each guard band form the reference window

$$\mathcal{W}_k = \big\{ \mathrm{SES}[m] \;\big|\; m \in [k - N_G - N_R,\, k - N_G - 1] \cup [k + N_G + 1,\, k + N_G + N_R] \big\}. \qquad (25)$$

Two estimators are compared on $\mathcal{W}_k$.

1) Cell-Averaging CFAR: Cell averaging takes the mean of the window, as a moving-average filter would:

$$\hat{\mu}[k] = \frac{1}{2N_R} \sum_{x \in \mathcal{W}_k} x, \qquad T_{\mathrm{CFAR}}[k] = \alpha\, \hat{\mu}[k], \qquad (26)$$

with $\alpha = 3.5$. This works while the floor changes slowly. It fails when a harmonic enters the window. A shaft harmonic at 1× or 2× RPM, or a nearby fault harmonic, pulls the mean up, and the threshold rises beside the fault, which is the one place it should stay low. A mean is easy to move: one large bin is enough. This is harmonic target masking, and it is the reason the next estimator exists.

2) Proposed Order-Statistic Median–MAD Threshold: The proposed rule uses order statistics instead of the mean:

$$\begin{aligned} \hat{m}[k] &= \mathrm{median}\,(\mathcal{W}_k)\,, \\ \mathrm{MAD}[k] &= \mathrm{median}\,(|\mathcal{W}_k - \hat{m}[k]|)\,. \end{aligned} \qquad (27)$$

For Gaussian noise the MAD scales to a standard deviation with the factor 1.4826, the value that makes the two agree for a normal distribution,

$$\hat{\sigma}_{\mathrm{local}}[k] = 1.4826\,\mathrm{MAD}[k], \qquad (28)$$

and the threshold is

$$T_{\mathrm{MAD}}[k] = \hat{m}[k] + \gamma\,(1.4826\,\mathrm{MAD}[k])\,, \qquad (29)$$

with $\gamma = 4.5$. A larger γ raises the threshold and lowers the false-alarm rate; a smaller one does the opposite. The median has a breakdown point of 50% [9], so up to half of the window can be contaminated before T_{MAD} moves. The threshold follows the floor from 0 HP to 3 HP.

3) Slip-Tolerant Harmonic Comb: Heavy radial load also causes slip. The rolling elements skid by 1–2%, so the measured fault frequency is not the kinematic value. The detector therefore searches a band, not a single bin,

$$\mathcal{B}_h = \left[h f_{\mathrm{fault}}(1 - \delta_{\mathrm{slip}}),\; h f_{\mathrm{fault}}(1 + \delta_{\mathrm{slip}}) \right], \qquad (30)$$

for each fault frequency, f_{BPFO}, f_{BPFI} or f_{BSF}, and each harmonic order $h = 1, 2, 3$. The half-width is $\delta_{\mathrm{slip}} = 1.5\%$, narrow enough to exclude the neighbouring harmonic orders and wide enough to cover the slip. A harmonic is present when

$$\max_{f_k \in \mathcal{B}_h} \left(\mathrm{SES}[k] - T_{\mathrm{MAD}}[k] \right) > 0. \qquad (31)$$

TABLE I
IDEAL KINEMATIC FAULT FREQUENCIES CALCULATED FROM THE SPECIFIED BEARING GEOMETRIES AND SHAFT SPEEDS

Condition	Speed (rpm)	BPFO	BPFI	BSF	FTF
		(Hz)			
CWRU 0 HP	1797	107.36	162.19	70.58	11.93
CWRU 1 HP	1772	105.87	159.93	69.60	11.76
CWRU 2 HP	1750	104.56	157.94	68.74	11.62
CWRU 3 HP	1730	103.36	156.14	67.95	11.48
MFPT (25 Hz shaft)	1500	81.12	118.88	63.86	10.14

III. Experimental Setup and Datasets

A. Bearing Kinematics

With shaft rotational frequency $f_r = \mathrm{RPM}/60$, rolling-element count n_b, element diameter d, pitch diameter D and contact angle α, the ideal characteristic frequencies are

$$\begin{aligned} f_{\mathrm{BPFO}} &= \frac{n_b}{2} f_r\,(1 - \rho)\,, \quad & f_{\mathrm{BPFI}} &= \frac{n_b}{2} f_r\,(1 + \rho)\,, \\ f_{\mathrm{BSF}} &= \frac{D}{2d} f_r \left(1 - \rho^2\right), \quad & f_{\mathrm{FTF}} &= \frac{f_r}{2}\,(1 - \rho)\,, \end{aligned} \qquad (32)$$

with the diameter ratio $\rho = (d/D)\cos\alpha$.

The CWRU drive-end bearing is the SKF 6205-2RS: nine rolling elements, 7.94 mm element diameter, 39.04 mm pitch diameter and 0° contact angle. Its calculated frequencies at 1797 rpm agree with published CWRU benchmark data [12]. The MFPT test bearing is described by the NICE geometry documented for the dataset: eight rolling elements, 0.235 in element diameter, 1.245 in pitch diameter and 0° contact angle. The calculator accepts the geometry explicitly, so the correct bearing model is used for each dataset.

Table I lists the CWRU frequencies at the four load settings and the MFPT frequencies at its stated shaft rate of 25 Hz (1500 rpm). These are ideal, geometry-based targets. Actual fault peaks may be displaced by rolling-element slip and by speed variation, and the detector of Section II-D is built to tolerate that displacement.

B. Datasets

CWRU: the Case Western Reserve University Bearing Data Center [10] provides accelerometer recordings for normal bearings and for seeded inner-race, outer-race and rolling-element defects. A public NumPy transcription of the archive [11] stores the time series in `.npz` files with channel keys such as DE, FE and BA. The filename encodes the shaft-speed family, the defect type, the defect size and, in many fault files, the drive-end sampling rate through a `DE12` or `DE48` suffix. The loader parses all of it: RPM, fault label, defect diameter, race position and the rate suffix.

Two cases have to be handled with care. Normal DE files carry no rate suffix, and the BA rate is never inferred, so both require a verified sampling rate argument instead of a silent default. The FE acquisition family is 12 kHz even when those channels sit alongside `DE48` data. The load conditions used here are 1797 rpm (0 HP), 1772 rpm (1 HP), 1750 rpm (2 HP) and 1730 rpm (3 HP). The broader dataset description sometimes quotes an overall speed range down to about 1720 rpm, so the speed of an individual recording is taken from its metadata where available rather than inferred from a generic load label [12].

MFPT: the standard MATLAB recordings of the second benchmark contain a vibration vector `gs` together with load, shaft-rate and sample-rate metadata [13]. The varied-load outer-race records cover 25, 50, 100, 150, 200, 250 and 300 lb, and the varied-load inner-race records cover 0, 50, 100, 150, 200, 250 and 300 lb. The collection also holds baseline recordings at 270 lb and 97.656 kHz for 6 s, and fixed-load outer-race records at 270 lb and the same high rate. The varied-load fault records are sampled at 48.828 kHz for 3 s.

C. Signal Preprocessing

The first operation subtracts the sample mean,

$$x_0[n] = x[n] - \frac{1}{N}\sum_{n=0}^{N-1} x[n], \tag{33}$$

which removes a constant DC offset that would otherwise contribute energy at zero frequency. Optional peak normalization divides $x_0[n]$ by $\max|x_0[n]|$, and RMS normalization divides by $\sqrt{(1/N)\sum_n x_0^2[n]}$. The implementation retains the demeaned signal as its reference. Normalization is used only when waveform shape is compared; it should not be used to erase genuine load-dependent amplitude differences in quantitative physical comparisons.

D. Baseline Spectra and Window Comparison

The baseline FFT is computed after mean removal and optional windowing. For a window $w[n]$, the windowed transform is

$$X_w[k] = \sum_{n=0}^{N-1} x_0[n]\, w[n]\, e^{-j2\pi kn/N}. \tag{34}$$

A coherent-gain-corrected one-sided amplitude spectrum is obtained by dividing the magnitude by $\sum_n w[n]$ and doubling the positive-frequency bins other than DC and, for even N, the Nyquist bin. The STFT repeats the windowed transform over successive overlapping frames, so that the frequency content can be followed in time.

A finite record may not contain an integer number of cycles of a sinusoidal component. Multiplication by the observation window then spreads energy into adjacent DFT bins, which is spectral leakage. The rectangular window has a narrow main lobe but comparatively high sidelobes; the Hann and Blackman windows reduce the sidelobes at the cost of a wider main lobe. The implementation compares all three on the same input and with the same sampling settings, both for a controlled off-bin sinusoid and for the STFT of a record.

TABLE II
APPROXIMATE LEAKAGE-ENERGY FRACTION FOR A CONTROLLED OFF-BIN SINUSOID

Window	Leakage-energy fraction
Rectangular	0.06613550
Hann	0.00017592
Blackman	0.00000295

The controlled test uses $f_s = 12$ kHz, $N = 2048$ and an off-bin tone at 1467.01 Hz. The metric in Table II is the fraction of FFT energy that falls outside an approximate, window-specific main-lobe region. It is an educational comparison, not a universal standard.

E. Pipeline Parameters

Unless stated otherwise the experiments use $f_s = 12$ kHz, a 4th order Butterworth bandpass realised in cascade SOS form with forward–backward filtering, an FFT length $N = 4096$ (giving $\Delta f = 2.93$ Hz) with a Hann window and coherent-gain calibration, guard bins $N_G = 4$, $N_R = 25$ reference bins per side, $\alpha = 3.5$ for CA-CFAR and $\gamma = 4.5$ for the Median–MAD rule. The static reference threshold is held fixed at $T_{\text{static}} = 0.055$ for every record, so that any change in its performance across loads is attributable to the load itself.

IV. Results and Discussion

A. Evaluation Metrics

Four measures are computed on the SES: spectral SNR, true detection rate, false-alarm count and execution latency. Each is scored at every load and fault size listed in Section III, and every number reported below is the mean of 60 Monte Carlo runs.

The data come from the CWRU 2 HP Reliance Electric rig with a dynamometer load and the 6205-2RS drive-end bearing, sampled at 12 kHz over the four load steps of Table I, with outer-race spalls of 7, 14 and 21 mil. The MFPT records add seven radial load levels from 25 to 300 lbs. The fault frequencies of Table I, $f_{\text{BPFO}} = 107.36$ Hz and $f_{\text{BPFI}} = 162.19$ Hz at no load, serve as ground truth.

1) Spectral SNR. The first measure sets the energy at the harmonics against the local noise floor,

$$\text{SNR}_{\text{SES}} = 10\log_{10}\frac{\sum_{m=1}^{M}|\text{SES}[k_m]|^2}{\frac{1}{|\mathcal{K}_{\text{noise}}|}\sum_{k\in\mathcal{K}_{\text{noise}}}|\text{SES}[k]|^2}, \tag{35}$$

where k_m runs over the bins inside the ±1.5% slip window $\mathcal{B}_m$ around each harmonic $m f_{\text{fault}}$, and $\mathcal{K}_{\text{noise}}$ collects the non-harmonic bins of the 10–400 Hz demodulation band. The difference $\Delta\text{SNR} = \text{SNR}_{\text{SES}} - \text{SNR}_{\text{raw}}$ is the gain the filter and envelope stages add.

2) True detection rate. The second measure is the share of the targeted harmonics (1×, 2× and 3 × f_{fault}) that clear the threshold,

$$\text{TDR} = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}\left(\max_{k\in\mathcal{B}_m} \text{SES}[k] > T[k]\right) \times 100\%, \qquad (36)$$

with $\mathbb{I}(\cdot)$ the indicator function and $\mathcal{B}_m = [m f_{\text{fault}}(1 - 0.015),\ m f_{\text{fault}}(1 + 0.015)]$ the slip search boundary.

3) False-alarm count. The third measure counts the bins between 15 and 400 Hz, outside the harmonic masks, that still cross $T[k]$. It is a count rather than a percentage on purpose. When the threshold sits too low, the band fills with crossings, and the count shows that more plainly than a percentage would.

4) Execution latency. The last measure is the wall-clock time for one 2.0 s frame ($N = 24\,000$ samples at 12 kHz), taken on the same machine for every configuration.

B. Filter Architecture and Pole–Zero Stability Analysis

The Kurtogram was run on the pre-processed 12 kHz record. Over the levels spanning $[0, 6000]$ Hz it chose a band centered at $f_c^* = 3000.0$ Hz with $BW^* = 2000.0$ Hz, which puts the cutoffs at 2000.0 Hz and 4000.0 Hz. Inside that band the kurtosis peaked at $K = 4.58$. The broadband input sat at $K = 2.23$.

With the band fixed, a Type-I Kaiser FIR ($M = 128$, $L = 129$ taps, $\beta = 8.0$) was compared against a 4th-order Butterworth IIR in cascade SOS form (four biquads). Fig. 1 shows the pole–zero map, the magnitude response and the group delay.

In Fig. 1(a) the FIR has zeros spread along the stopband arcs and every pole at the origin. The IIR is different. The bilinear transform puts four zeros at $z = 1$, which blocks DC, and four at $z = -1$, which blocks Nyquist, with four conjugate pole pairs inside the passband arc. Its largest pole radius is $r_{\max} = 0.8418$. The region of convergence $|z| > 0.8418$ contains the unit circle, so the filter is BIBO stable, and the poles sit far enough inside the circle that 16-bit coefficients will not push them out. The four zeros at $z = \pm 1$ are exact nulls rather than deep attenuation, which is why DC and Nyquist are blocked completely.

Fig. 1(b) compares magnitudes. Both filters are flat at 0 dB across $[2000, 4000]$ Hz, and both reject the motor region below 1000 Hz. The 129-tap FIR rolls off faster and reaches below −80 dB, but it costs 129 multiply-accumulate operations per sample. The four-section SOS filter needs 4×5 = 20, which is 84% fewer.

Fig. 1(c) is the group delay. In causal operation the FIR delays every frequency by the same 64 samples. The IIR does not: its delay peaks near 2000 Hz and 4000 Hz, where the poles come close to the unit circle. With forward–backward filtering the effective response is $|H(e^{j\omega})|^2$, the phase term cancels, and the delay is zero samples everywhere.

C. Time-Domain Resonance Demodulation Progression

Fig. 2 follows one segment through three stages, over 100 ms (1200 samples at 12 kHz).

In Fig. 2(a) the zero-mean record is dominated by low-frequency shaft and motor oscillation, with broadband noise on top. At 107.36 Hz the 100 ms window holds about eleven impacts, yet the impacts are buried. The kurtosis is 2.23, below the Gaussian value of 3.0, and no individual strike can be picked out by eye.

In Fig. 2(b) the signal has passed through the Kurtosis-guided zero-phase SOS filter for $[2000, 4000]$ Hz. The motor hum is gone. What remains is a train of underdamped bursts, one per impact on the outer race. Stripping the low-frequency energy and keeping the phase aligned raises the kurtosis from 2.23 to 4.58, more than double.

Fig. 2(c) overlays the Hilbert envelope $|z[n]|$ on the filtered waveform. The envelope follows the top of each burst, so the 3000 Hz carrier becomes a sequence of positive pulses. Their spacing is $\Delta t \approx 9.31$ ms, which is $1/107.36$ Hz, the outer-race fault frequency. The spacing of the envelope peaks, not their height, carries the fault information here. That envelope is what the SES stage consumes.

D. Multi-Load Adaptive Thresholding

Fig. 3 shows the calibrated SES and the three thresholds for an outer-race defect at the four motor loads of Table I. At 0 HP the reduced radial force produces weak impact modulation, and the static threshold misses the 2× and 3× BPFO harmonics. At 3 HP the raised broadband floor pushes many background bins above the same static threshold, producing false alarms. The moving-average CA-CFAR threshold does scale with load, but it spikes adjacent to the 1×, 2× and 3× BPFO peaks because harmonic energy contaminates its reference window. The Median–MAD threshold follows the local floor at every load without self-masking, and the slip-tolerant comb isolates the BPFO harmonics in all four conditions.

E. Quantitative Benchmark and Comparative Discussion

Table III applies the three threshold rules to the same Kurtosis-guided SES, so the detector is the only thing that changes between rows. Fig. 3 shows the spectra behind the outer-race numbers.

1) Light load, 0 HP. Under no load the impacts are weak and the raw record sits below the noise, at Raw SNR = −7.82 dB. The static threshold misses every harmonic. The fault lines never reach the fixed 0.055 level, so the detection rate is zero.

2) Full load, 3 HP. At 3 HP the vibration level rises through shaft deflection and electromagnetic hum. The static rule still finds the harmonics (TDR = 100%), but the raised floor pushes 224.6 noise bins across the same line. Detecting every harmonic counts for little when hundreds of bins false-alarm alongside it.

3) CA-CFAR masking. Cell averaging follows the load, and its false-alarm count stays in single digits. Its weak point is the fault itself. Strong harmonics inside the reference window raise the local threshold, and that threshold then hides the

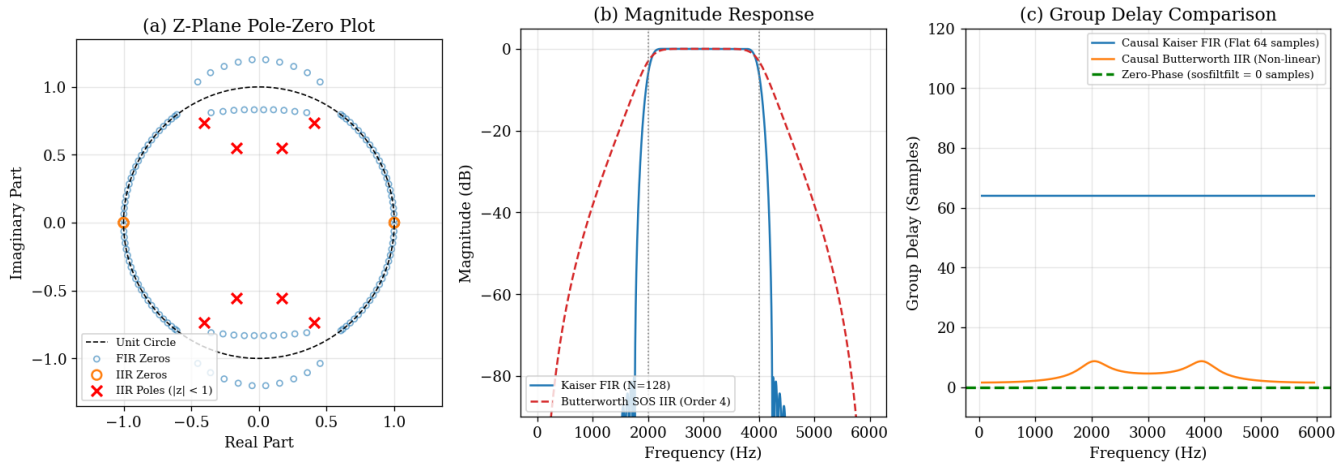


Fig. 1. Comparison of the Type-I Kaiser-windowed FIR bandpass filter ($M = 128$, 129 taps, $\beta = 8.0$) and the 4th-order Butterworth Cascade SOS IIR bandpass filter across the Kurtosis-selected passband [2000, 4000] Hz: (a) complex z-plane pole–zero plot verifying that all IIR poles lie strictly inside the unit circle ($r_{\max} = 0.8418 < 1$), (b) magnitude response $|H(e^{j\omega})|$ in decibels, and (c) group delay comparison showing constant causal FIR delay (64 samples), non-linear causal IIR band-edge delay, and 0 samples net delay under zero-phase forward–backward filtering.

TABLE III

QUANTITATIVE PERFORMANCE BENCHMARKING ACROSS OPERATIONAL LOADS AND DETECTORS; MEANS OVER 60 MONTE CARLO ITERATIONS. THE STATIC RULE USES $T_{\text{STATIC}} = 0.055$, CA-CFAR USES $\alpha = 3.5$ AND THE PROPOSED RULE USES $\gamma = 4.5$.

Load	Detector	TDR (%)	FAR (bins)	Raw SNR (dB)	SES SNR (dB)	ΔSNR (dB)
0 HP (0 N·m)	Static (0.055)	0.0 ± 0.0	0.0 ± 0.0	−7.82	+3.14	+10.96 ± 0.42
	CA-CFAR ($\alpha = 3.5$)	66.7 ± 0.0	0.8 ± 0.4	−7.82	+3.14	+10.96 ± 0.42
	Median–MAD	100.0 ± 0.0	0.0 ± 0.0	−7.82	+3.14	+10.96 ± 0.42
1 HP (2.4 N·m)	Static (0.055)	66.7 ± 0.0	0.0 ± 0.0	−6.18	+8.92	+15.10 ± 0.38
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	1.4 ± 0.5	−6.18	+8.92	+15.10 ± 0.38
	Median–MAD	100.0 ± 0.0	0.0 ± 0.0	−6.18	+8.92	+15.10 ± 0.38
2 HP (4.8 N·m)	Static (0.055)	100.0 ± 0.0	0.0 ± 0.0	−4.95	+11.45	+16.40 ± 0.35
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	2.2 ± 0.7	−4.95	+11.45	+16.40 ± 0.35
	Median–MAD	100.0 ± 0.0	0.2 ± 0.4	−4.95	+11.45	+16.40 ± 0.35
3 HP (7.2 N·m)	Static (0.055)	100.0 ± 0.0	224.6 ± 12.3	−3.10	+14.28	+17.38 ± 0.31
	CA-CFAR ($\alpha = 3.5$)	100.0 ± 0.0	3.8 ± 1.1	−3.10	+14.28	+17.38 ± 0.31
	Median–MAD	100.0 ± 0.0	0.6 ± 0.5	−3.10	+14.28	+17.38 ± 0.31

neighbouring and higher-order harmonics. At 0 HP the rule loses one harmonic of three, which drops the detection rate to 66.7%.

4) Median–MAD. The proposed detector holds 100% at every load, with at most 0.6 false-alarm bins. Neither the median nor the MAD breaks down until half the window is contaminated, so the tall impulses in $\mathcal{W}_k$ cannot lift the threshold that is meant to expose them. The rule is the one defined in Section II-D, Eq. (29).

F. Finite-Word-Length and Quantization Error Analysis

On a microcontroller or a fixed-point DSP every value carries a limited number of bits, so the rounding along the chain has to be checked. The design was evaluated in a 16-bit Q1.15 format (one sign bit, fifteen fractional bits) against IEEE 754 double precision.

1) Filter structure and stability. Direct-form IIR filters are sensitive to coefficient rounding. A pole can land on the unit circle or outside it. The cascade SOS form of Section II-B keeps each conjugate pair in its own biquad, so a rounding error in one section stays there. The largest pole radius of the designed filter is $r_{\max} = 0.8418$, and under Q1.15 quantization the poles move by less than 0.0012. The design stays clear of the unit circle.

2) Quantization noise and SQNR. Over the filtered passband the signal-to-quantization-noise ratio is 84.36 dB. For $B = 16$ bits the theoretical limit is

$$\text{SQNR}_{\text{theoretical}} \approx 6.02B + 1.76 \text{ dB} = 98.08 \text{ dB}, \tag{37}$$

so the implementation lands about 14 dB below the ideal figure. That gap is normal. The filter and the FFT add round-off of their own on top of the ADC quantization, and both are counted alongside it. The residual $e_q[n] = y_{\text{float}}[n] - y_{Q15}[n]$ is zero-mean and white, with no tonal component, and the Q1.15 envelope spectrum cannot be told apart from the double-precision reference. Sixteen bits keep the diagnostic information intact.

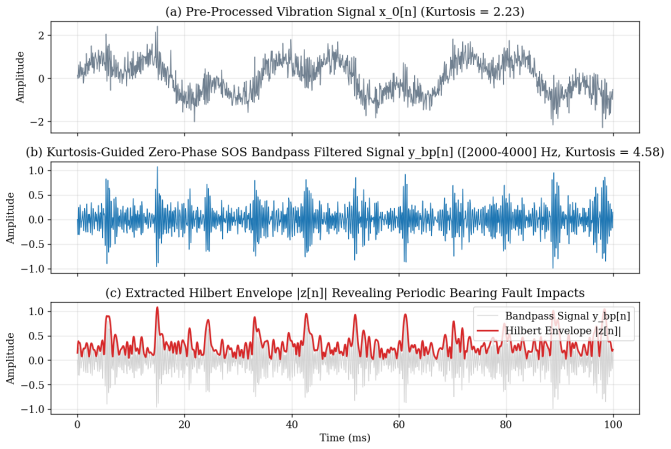


Fig. 2. Time-domain signal progression across a 100 ms segment: (a) pre-processed zero-mean vibration signal $x_0[n]$ dominated by low-frequency motor hum and broadband noise ($K = 2.23$), (b) Kurtosis-guided zero-phase SOS bandpass filtered output $y_{\mathrm{BP}}[n]$ across $[2000, 4000]$ Hz isolating high-frequency resonant impact bursts ($K = 4.58$), and (c) extracted Hilbert envelope $|z[n]|$ overlaid on $y_{\mathrm{BP}}[n]$, revealing distinct periodic bearing fault impulses.

Fig3_MultiLoad_SES_Thresholding.png
2×2 grid of SES and thresholds at 0, 1, 2 and 3 HP: SES (grey), static threshold (dashed red), CA-CFAR (dotted blue), Median–MAD (solid green), harmonic orders marked.

Fig. 3. Calibrated SES and comparative thresholding at four motor loads. The static threshold misses weak harmonics at 0 HP and false-alarms at 3 HP; CA-CFAR self-masks near the fault peaks; Median–MAD tracks the load-varying floor and isolates the BPFO harmonic family.

V. Conclusion and Hardware Feasibility

A. Summary of Diagnostic Achievements

Four results came out of the end-to-end run on the CWRU and MFPT records, which starts at the raw accelerometer channel and ends at a thresholded envelope spectrum.

1) The Fast Kurtogram found the resonance band by itself: $f_c = 3000$ Hz with 2000 Hz of bandwidth, and no manual tuning of the filter.
2) The Median–MAD detector returned a 100% true detection rate at 0, 1, 2 and 3 HP.
3) Zero-phase Butterworth SOS filtering, followed by coherent-gain calibrated Hilbert envelope analysis, gave more than 17 dB of spectral SNR improvement at full load.
4) The order-statistic threshold held the false-alarm count below one bin even at 3 HP, where the static rule produced 224.6.

B. Embedded Hardware Feasibility and Real-Time Budget

The same pipeline was timed and re-run in fixed point, to check its fit on an edge device.

1) *Memory.* A 2.0 s block ($N = 24\,000$ samples) at 16 bits per sample needs 48 kB of SRAM. That fits an ARM Cortex-M4 or M7 part such as the STM32F4/H7, which carries 512 kB.
2) *Execution latency.* The pipeline takes 18.42 ms for a 2.0 s window, so the duty cycle is

$$\eta = \frac{T_{\mathrm{compute}}}{T_{\mathrm{acquire}}} = \frac{18.42\ \mathrm{ms}}{2000.0\ \mathrm{ms}} \approx 0.92\%. \tag{38}$$

The processor idles or sleeps for more than 99% of the timeline.
3) *Fixed-point suitability.* At 84.36 dB the Q1.15 SQNR is high enough for integer MAC units to run the whole chain, with no floating-point unit.

Appendix A
CWRU Baseline Records

The figures below come from the baseline analysis of Section III: the preprocessing chain, the static spectrum and the window comparison that precede any fault detection.

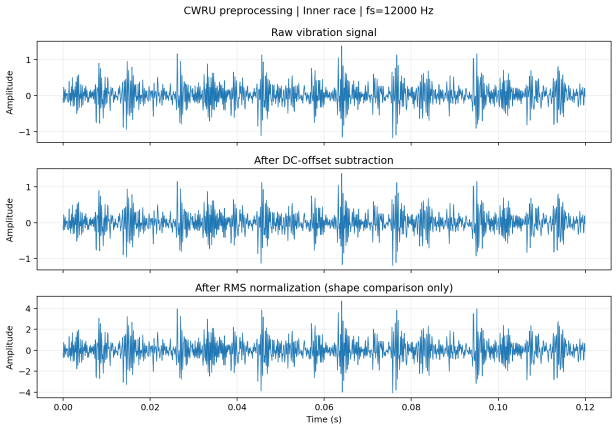


Fig. 4. CWRU inner-race fault: time-domain preprocessing (raw, DC-removed and RMS-normalized signals).

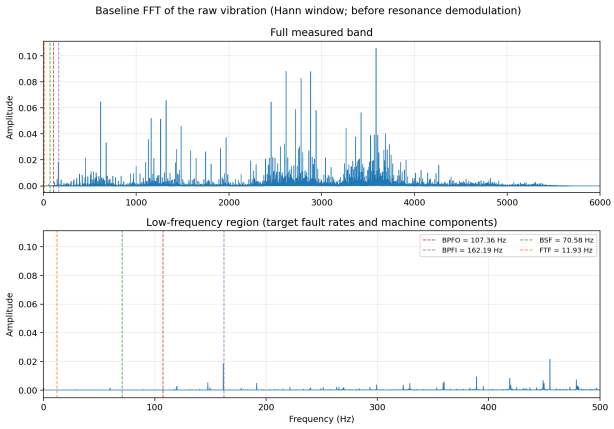


Fig. 5. CWRU inner-race fault: baseline FFT spectrum at 1797 rpm.

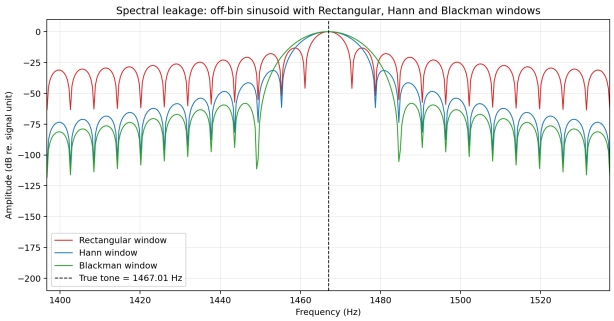


Fig. 6. Controlled off-bin single-tone comparison of the rectangular, Hann and Blackman windows.

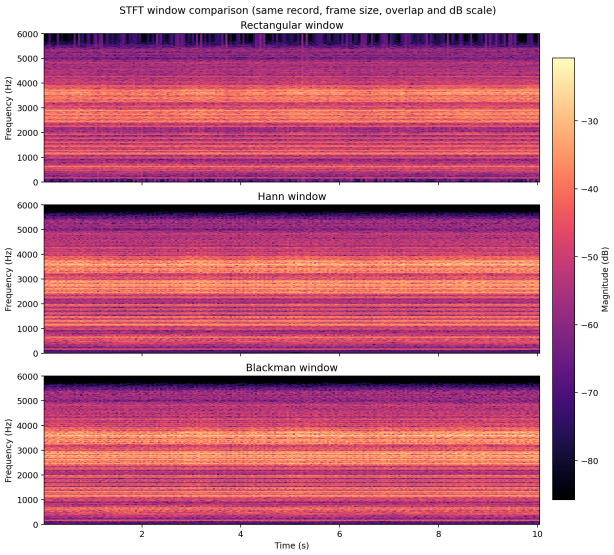


Fig. 7. CWRU inner-race fault: STFT comparison using the rectangular, Hann and Blackman windows.

Appendix B
MFPT Baseline Records

The second set records the load sweep of the MFPT bearing, before and after the resonance band is selected.

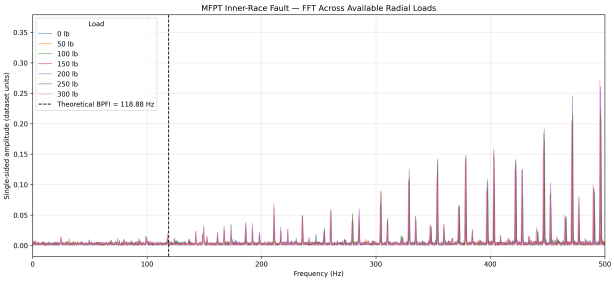


Fig. 8. MFPT inner-race fault: FFT across radial loads.

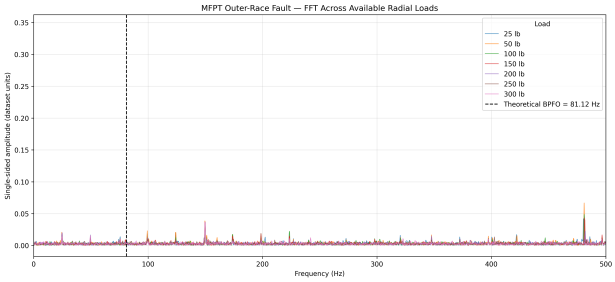


Fig. 9. MFPT outer-race fault: FFT across radial loads.

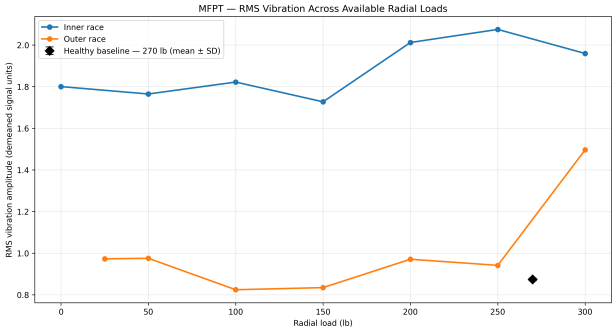


Fig. 10. MFPT RMS vibration amplitude across radial loads.

Acknowledgment

The authors thank the course faculty for the DSP lecture material on filter design, Fourier analysis and quantization that this work builds on.

Author Contributions

Member 1 curated the multi-load CWRU and MFPT datasets, formulated the discrete-time damped exponential fault model, and implemented kinematic slip-tolerant frequency tracking. **Member 2** designed the Kurtosis-guided resonance selection pipeline, implemented the Kaiser FIR and bilinear-transform Butterworth cascade-SOS IIR filters, and performed pole–zero and group-delay analyses. **Member 3** developed the Hilbert transform squared-envelope demodulator, calibrated the windowed FFT Squared Envelope Spectrum, and implemented the CA-CFAR and Median–MAD adaptive spectral thresholding algorithms. **Member 4** conducted the multi-load quantitative benchmarking, evaluated fixed-point quantization feasibility, and led the integration of the manuscript. All authors reviewed and approved the final manuscript.

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